Lecture 1 Homework

Programming

1. Write and test code for the Euclidean Algorithm.
2. Write and test code that will find a couple of integers, x_0 and y_0, for given integers m and n, such that $x_0 m + y_0 n$ will yield the smallest positive number. This is, of course, the Extended Euclidean Algorithm.

Exercises

1. Prove directly, from the very definition of divisibility, that if $d|x$ and $d|y$, then $d|x-y$ and $d|x+y$.
2. Show that for every n, $n+1$ and n are relatively prime.
3. Show that for every n: n and n^2+1 are relatively prime.
4. Factor 1028 into primes.
5. Factor 2057 into primes.
6. Find a proper factor of $111{,}111{,}111{,}111{,}111$.
7. Find a proper factor of $110{,}111{,}101{,}111{,}011$.
8. Find a proper factor of $101{,}010{,}101{,}010{,}101$.
9. Explain why n^2-1 cannot be prime for any $n>2$.
10. Explain why 3^n-1 cannot possibly be a prime number if $n>1$.
11. Find $\gcd(102313, 103927)$ without factoring.
12. Based on the previous problem find the inverse of $\overline{102{,}313}$ in $Z^{\times}_{103927}$
13. For an integer n, show that the greatest common divisor of the integers n^3+n^2+n+1 and n^2+n+1 is just 1.
14. For an integer $n>0$, what is greatest common divisor of n^3+n^2+n+1 and $n^8+n^7+n^6+n^5+n^4+n^3+n^2+n+1$?
15. By trial and error, find a multiplicative inverse to $3 \bmod 7$.
16. Find the multiplicative inverse to $5 \bmod 7$.
17. Find the multiplicative inverse to $7 \bmod 11$.
18. Find the multiplicative inverse to $5 \bmod 13$.
19. Find the multiplicative inverse to $9 \bmod 11$.
20. Find the multiplicative inverse to $n \bmod 2n-1$.
21. Find the multiplicative inverse to $n \bmod 2n+1$.
22. Find the multiplicative inverse to $n \bmod 3n+1$.
23. Find the multiplicative inverse to $n \bmod n^2+1$.
24. Find the multiplicative inverse to $n \bmod n^2-1$.

25. Find the multiplicative inverse of 21 in $Z_{25}^{\times}$.
26. Find an integer x so that $(x\%1009)\%1013 \neq x\%1013$.
27. Take two positive integers n and N with n not dividing N. Find an integer x so that $(x\%N)\%n \neq x\%n$.
28. How many elements does the set Z_n have?
29. How many elements does the set $Z_{30}^{\times}$ have?
30. Compute $2^{1000}\%11$.
31. Compute $3^{1000}\%11$.
32. Find three distinct classes $x \bmod 105$, so that $x^2 \equiv 1 \bmod 105$.
33. Find three distinct classes $x \bmod 143$, so that $x^2 \equiv 1 \bmod 143$.

Suggested Reading

- Garrett – Section 1.2, Chapter 7
- Smart – Section 1.1, 1.3.1, A.1, A.2
- Forouzan – Chapter 2, Section 9.3 to page 270)